

Problem Set 10 — Circular Motion & Gravitation

Physics 2026-27 · due at the start of class, Monday, December 7 · Unit 3 Test Thu, Dec 10

NAME _____

DATE _____

How to work this set. Equations this week: $a_c = v^2/r$ (always toward the center), $v = 2\pi r/T$, $F_c = mv^2/r$, and $F_g = Gm_1m_2/r^2$ with $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$. Remember: "centripetal force" is a *label* for whatever real force points centerward — it never gets its own arrow on an FBD. Scientific notation is back; use $\approx$ honestly.

1 Constant speed, still accelerating

CONCEPT

- A car circles a roundabout at a perfectly steady 20 km/h. Its speedometer never moves — yet it is accelerating. Explain how both can be true.
- True or false, with one sentence: a passenger in that car is pushed outward by a "centrifugal force."

2 Straight to the formula

 A_c

A cyclist rounds a curve at 9.0 m/s with a turning radius of 4.5 m . Run KDEA for the centripetal acceleration — magnitude and direction.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	$A_c = v^2/r$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

3 Ball on a string

PERIOD → FORCE

A 0.20 kg ball swings in a horizontal circle of radius 0.75 m , completing one lap every 0.60 s .

- Find its speed. ($v = 2\pi r/T$)
- Find its centripetal acceleration.
- Find the net centripetal force — and name the real force that supplies it.

K	EVERY GIVEN, WITH UNITS
D	THREE TARGETS, THREE UNITS
E	$V = 2\pi R/T$ · $A_C = V^2/R$ · $F_C = MA_C$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY · $\approx$ WHERE π FORCES IT

4 The flat curve

FRICTION AS F_C

An 1100 kg car takes a flat curve of radius 45 m at 14 m/s .

- How much centripetal force does the turn require?
- What minimum coefficient of static friction keeps the car on the road? (Set $\mu_s mg = mv^2/r$ and watch the mass cancel.)
- Why μ_s and not μ_k ? One sentence — a Workshop I subscript question, three months later.

K	EVERY GIVEN, WITH UNITS
D	THE TARGETS & THEIR UNITS (M HAS NONE — SAY SO)
E	$F_C = MV^2/R$ · $M_s = V^2/(RG)$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

5 Inverse-square reasoning

GRAVITATION

No calculator for this one — pure proportional reasoning from $F_g = Gm_1m_2/r^2$:

- The distance between two masses doubles. What happens to F_g ?
- The distance shrinks to one-third of the original. What happens to F_g ?
- Both masses double *and* the distance doubles. Now what?

REASONING

6 Earth pulls the satellite

$$F_G = GM_1M_2/R^2$$

A 600 kg satellite orbits at $r = 7.0 \times 10^6$ m from Earth's center. ($M_{\text{earth}} = 5.97 \times 10^{24}$ kg.) Run KDEA for the gravitational force on it — this is the scientific-notation workout Unit 0 trained you for, so track the exponents in your A row.

K	EVERY GIVEN, WITH UNITS, IN SCIENTIFIC NOTATION
D	THE TARGET & ITS UNIT
E	$F_G = GMM/R^2$
A	SYMBOLS FIRST · EXPONENT BOOKKEEPING SHOWN

7 Name the centripetal force

SOURCES

For each, name the *real* force (or forces) playing the centripetal role — "centripetal force" is not an acceptable answer:

- The Moon orbiting Earth.
- A coin resting on a spinning turntable.
- A roller-coaster car at the very top of a vertical loop. (Two forces — both point the same way there.)
- A car on a banked, frictionless curve.

ANSWERS

Graded on the homework 4-3-2-1 scale. Last problem set of the semester's new content — the **Unit 3 Test (Thu, Dec 10)** covers this material plus orbits, and midterms (Units 0-3) follow the week after.